Band-1 Hadamard order 14684 – Classical Paley-I (q=14683); IN SPK via Paley-I; Gap 2052 Still OPEN

H_14684 Paley-I PASS; Paley-II FAIL/SKIP (q=7341=3*2447 NOT prime; ===3 mod 4 needs ===1); Kronecker FAIL/SKIP (4*3671; 3671odd prime ===3 mod 4 NOT Hadamard-admissible; 3671 odd prime ===3 NOT Hadamard-admissible -> FAIL/SKIP; 3671=3671 (odd prime)); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-14

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-13_Band1_Hadamard_14668_PaleyII` (current bookkeeping); `2026-10-13_Band1_Hadamard_14` `2026-10-13_Band1_Hadamard_14640_PaleyI` (Paley-I construction template); `2026-10-13_Band1_Hadamard_1462` `2026-10-13_Band1_Hadamard_14620_PaleyII` (shape-class peer ===12/m===3); **Lab.** `/workspace/experiments/band1/hadamard_14684_paleyI/`.
Cutoff. 2026-10-14T05:20:22Z (~1:20AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14684 only (verified max|HH^T-nI|=0), via classical Paley-I q=14683 (prime ===3 mod 4). 14684 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14684: Paley-I **PASS** ($q = 14683$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=7341=3*2447 NOT a prime power; ===1 mod 4 but composite); Kronecker **FAIL/SKIP** ($14684 = 4 \cdot 3671$; $3671 \equiv 1 \pmod 4$ odd NOT Hadamard-admissible; $3671 = 2^2 \cdot 3 \cdot 5 \cdot 61$; do NOT invent H_3671). **14684** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture; 14684 is a long-known Hadamard order and this is a finite on-box verification, NOT a new-order claim. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14628+14620+14596+14592+14564+14552+14544+14520+14508+14504+14480+14476+14460+144** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **14684**.

- H_{14684} = Paley-I ($q = 14683$) – classical SPK
- Paley-II for 14684: **FAIL/SKIP** ($q = 7341 = 13 \cdot 563$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 14684: **FAIL/SKIP** (4·3671; $3671 \equiv 1 \pmod 4$ odd NOT Hadamard-admissible; $3671 = 2^2 \cdot 3 \cdot 5 \cdot 61$)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_14684.py
```

Results. overall_pass=True; units 847/847; row blocks 115/115 (block size 128); wall 9774.2839s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-14T05:20:22Z (~1:20AM ET)

Arithmetic

- Target $n = 14684 = q + 1$ with $q = 14683$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 7341 = 13 \cdot 563$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $14684 = 4 \cdot 3671$; $3671 \equiv 1 \pmod 4$ odd NOT Hadamard-admissible; $3671 = 2^2 \cdot 3 \cdot 5 \cdot 61$ — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14684 Paley-I	Paley-I q=14683	0	9736.6281
H_14684 Paley-II	FAIL/SKIP (q=7341=3*2447 NOT prime; ===3 mod 4 needs ===1)	–	–

Object	Method	max|HHˆT-nI|	Wall (s)
H_14684 Kronecker	FAIL/SKIP (4*3671; 3671odd prime ===3 mod 4 NOT Hadamard-admissible; 3671 odd prime ===3 NOT Hadamard-admissible -> FAIL/SKIP; 3671=3671 (odd prime))	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
14684 in SPK	**True** (Paley-I q=14683)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false

- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)
- 14684 is a long-known Hadamard order (Paley 1933 family); finite on-box verification only, NOT a new-order claim

Next optional

NEXT direct route after 14684 (FIRST any direct = Paley-II): q2=7349->n=14700 Paley-II DIRECT (prime ===1 mod 4; DUAL route: Paley-I q=14699 also eligible; Kronecker 14700=4*3675; 3675=3*5^2*7^2 odd NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14668_paleyII or hadamard_14684_paleyI) | Kronecker-only NOTE 14688=4*3672 (H_3672 via Paley-I q=3671; no direct Paley route; note-only, unbuilt) | Kronecker-only NOTE 14576=4*3644 (H_3644 via Paley-I q=3643; below 14684, note-only, unbuilt) | skipped Paley-II DIRECT orders below 14684 still unbuilt (template hadamard_14668_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)